

Experiments: Study of LAN Topology, Network Device and Address Configuration

Problem Statement

1. Draw the Network Topology:

- Draw and identify the LAN topology of your organisation. And label all components such as Router, Switch, Access Point, and end devices.

→ Below is the detail case study of Campus Network.

- Layer 3 Core Switches in the Server Room
- Managed by POE + Switch
- Wireless Access Point (APs)
- Multiple Ethernet port in Class & Laboratory.

→ Network Topology

Hybrid Hierarchical Topology which Combines

- Tree topology
- Extended Star topology
- Ring topology [Server Room].

→ Network hierarchical layer

1) Core layer - It contains Server Room and Server Firewall, Routers, 2 - L3 + Switches, Patch Rack Server, Inverter, etc.

2) Distribution layer - Here it contains 4 Switches
following are the 4 - Switches are connected

- ① Main Build → R7 → (2nd Floor)
 ② Main Build → R1 → (3rd Floor)
 ③ Main Build → R6 → (1st Floor)
 ④ Main Build → R2 → (2nd Floor)

And, each - 4 switches containing →

① 18-port Gigabit Easy Smart Switch with 16-port. TL-SG1218MPE

② SG3428XMP - 24-port Gigabit & 4-port 10 GbE SFP+, L2+ Managed. Switches with 24 ports

Here is the detail discussion and use of SG1218MPE and SG3428XMP

* SG1218MPE - As we know 18-port Gigabit "Easy Smart" Switch, which support 16 ports PoE+.

Example:- It is used for CCTV Camera, wifi access point, etc.

* SG3428XMP - Well, it is 24-PoE+ port Uplinks, 10 Gb SFP, it is usually act as main distribution point for each floor. Higher-end managed Layer 2+ Switch

→ Access layer: It consists of 3 switches

(i) R1 [Room-301, 3rd Floor]

(ii) R2 [Room-05, Ground Floor]

(iii) R3 [Room-10, Ground Floor]

↳ All the switches act as access layer which help to provide network connectivity to all class room, lab, faculty rooms, etc.

It is connected directly to :-

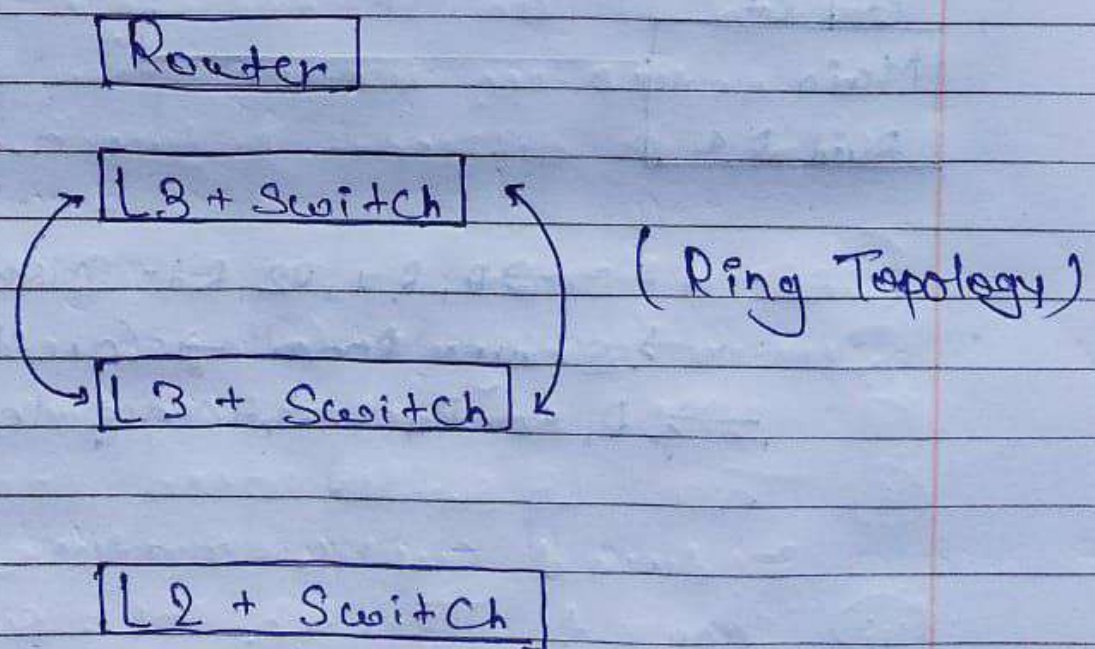
(i) Ethernet Port

(ii) Wireless APs

(iii) Lab System

(iv) Outdoor wireless AP (etc.).

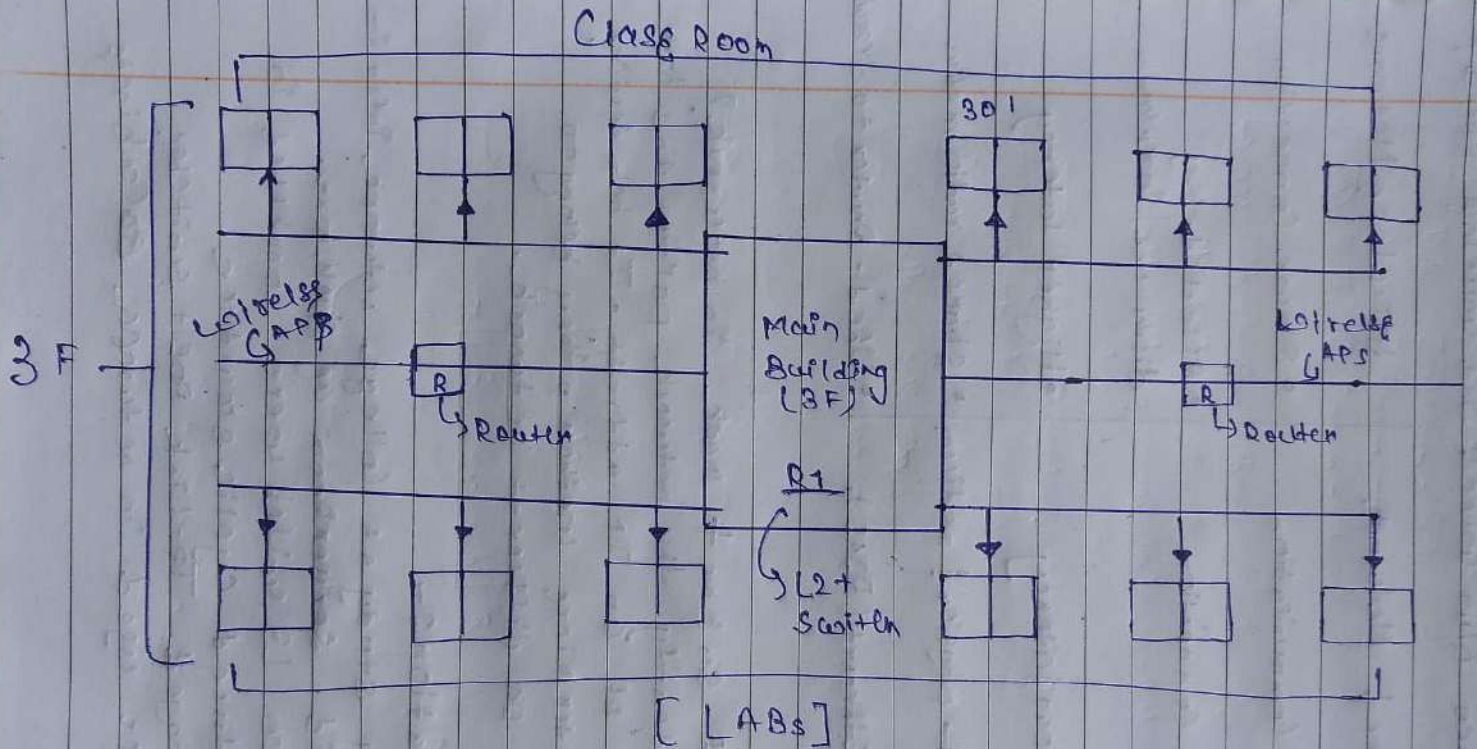
→ Server Room Architecture / Topology



Topology Design / Structure



- R, R7, R2, R6 - Distribution Layer
- Server Room → Core Layer
- R1 - R7 → Access Layer



It represent ethernet port for each class & lab (All are wired port)

Ques → Difference b/w wired and wireless speed?

Wired Network	Wireless Networks
• Faster Communication	• Slower Connection
• Stable Connection	• Affected by distance / Interference
• Very-low latency	• Higher latency
• Direct physical medium	• Shared radio medium
Used - Ethernet (RJ45).	• Used - Router to distrib.

Ques → Role of subnet mask in network?

It identifies the network portion and host portion of an IP address. It helps devices to determine whether they belong to same or different network.

Observations:-

During study of College network infra:-

- (i) Used Server Room as backbone.
- (ii) Two L3 switches were connected in reduced cyclic structure
- (iii) Managed TP-Link PoE + switches were used in network cabinets

Advantage of Existing Network distribution:-

- Scalable architecture
- Centralized managed
- Support PoE+devices
- Easy expansion of floors & class room
- Support both wired and wireless network
- Redundant link for failover.

Conclusion:-

The observed campus LAN follows a hybrid hierarchical topology consisting of tree, Extended Star Ring. The network used managed PoE Switches, wireless Aps, Ethernet ports & Centralized Server infrastructure to provide efficient wired & wireless communication, across the Campus.

>MACAddress
>IP Address (IPv4/6)
>Subnet Mask
Command used: ipconfig

```
Ethernet adapter Ethernet:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Ethernet adapter Ethernet 2:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Ethernet adapter vEthernet (Default Switch):  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::18a3:5d8f:ac77:9f1d%14  
    IPv4 Address. . . . . : 10.10.0.1  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
  
Wireless LAN adapter Local Area Connection* 8:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Wireless LAN adapter Local Area Connection* 9:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Wireless LAN adapter Wi-Fi:  
    Connection-specific DNS Suffix  . : home.network  
    IPv6 Address. . . . . : 2405:201:4012:7a3b:ed71:5f7a:9c21:8b23  
    Temporary IPv6 Address. . . . . : 2405:201:4012:7a3b:b4d2:02e6:6a8f:cd11  
    Link-local IPv6 Address . . . . . : fe80::e5a1:3c23:7d8f:12b4%21  
    IPv4 Address. . . . . : 172.16.45.78  
    Subnet Mask . . . . . : 255.255.240.0  
    Default Gateway . . . . . : 172.16.32.1  
    DHCP Server . . . . . : 172.16.32.1  
    DNS Servers . . . . . : 1.1.1.1  
                           1.0.0.1  
    NetBIOS over Tcpip. . . . . : Enabled  
  
Ethernet adapter Bluetooth Network Connection:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix  . :  
  
Ethernet adapter vEthernet (WSL (Hyper-V firewall)):  
    Connection-specific DNS Suffix  . :  
    Link-local IPv6 Address . . . . . : fe80::c6b7:2e5d:6f77:2a1d%32  
    IPv4 Address. . . . . : 172.28.128.1  
    Subnet Mask . . . . . : 255.255.240.0  
    Default Gateway . . . . . :
```


Ethernet

Command used: netsh wlan show interfaces

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\kamal> netsh wlan show interfaces

There is 1 interface on the system:

Name                           : Ethernet
Description                     : Intel(R) Ethernet Connection (7) I219-V
GUID                            : 8d3f7b4e-2a9f-4d1c-bc7e-5d1f2e9b6a3c
Physical address                : 1C:1B:0D:4E:5F:60
Interface type                  : Dedicated
State                           : connected
Media State                     : connected
Connection-specific DNS Suffix : corp.example.com
IPv4 Address                    : 10.10.20.15
Subnet Mask                     : 255.255.255.0
Default Gateway                 : 10.10.20.1
IPv6 Address                    : 2001:db8:abcd:1234:1c1b:d0ff:fe4e:5f60
Link-local IPv6 Address        : fe80::1c1b:d0ff:fe4e:5f60%12
DNS Servers                     : 10.10.20.5
                                10.10.20.6
DHCP Enabled                    : Yes
DHCP Server                     : 10.10.20.2
Lease Obtained                  : 24 May 2025 10:15:42
Lease Expires                   : 25 May 2025 10:15:42
Speed                           : 1 Gbps
Duplex                          : Full Duplex
Auto Configuration              : Enabled

PS C:\Users\kamal> █
```


Wi-Fi

Command used: netsh wlan show interfaces

```
Windows PowerShell
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PS C:\Users\kamal> netsh wlan show interfaces

There is 1 interface on the system:

    Name                : Wi-Fi
    Description          : Intel(R) Wi-Fi 6 AX201 160MHz
    GUID                 : b2f6e4d1-2b8f-4f11-9d73-7c9a4f2a88c1
    Physical address     : 8c:4d:79:2a:b3:11
    State                : connected
    SSID                 : GalaxyFiber_5G
    BSSID                : 34:60:f9:ab:72:91
    Network type         : Infrastructure
    Radio type           : 802.11ax
    Authentication       : WPA2-Personal
    Cipher               : CCMP
    Connection mode      : Auto Connect
    Channel              : 149
    Receive rate (Mbps)  : 866.7
    Transmit rate (Mbps) : 866.7
    Signal               : 94%
    Profile               : GalaxyFiber_5G

    Hosted network status : Not available

PS C:\Users\kamal> █
```